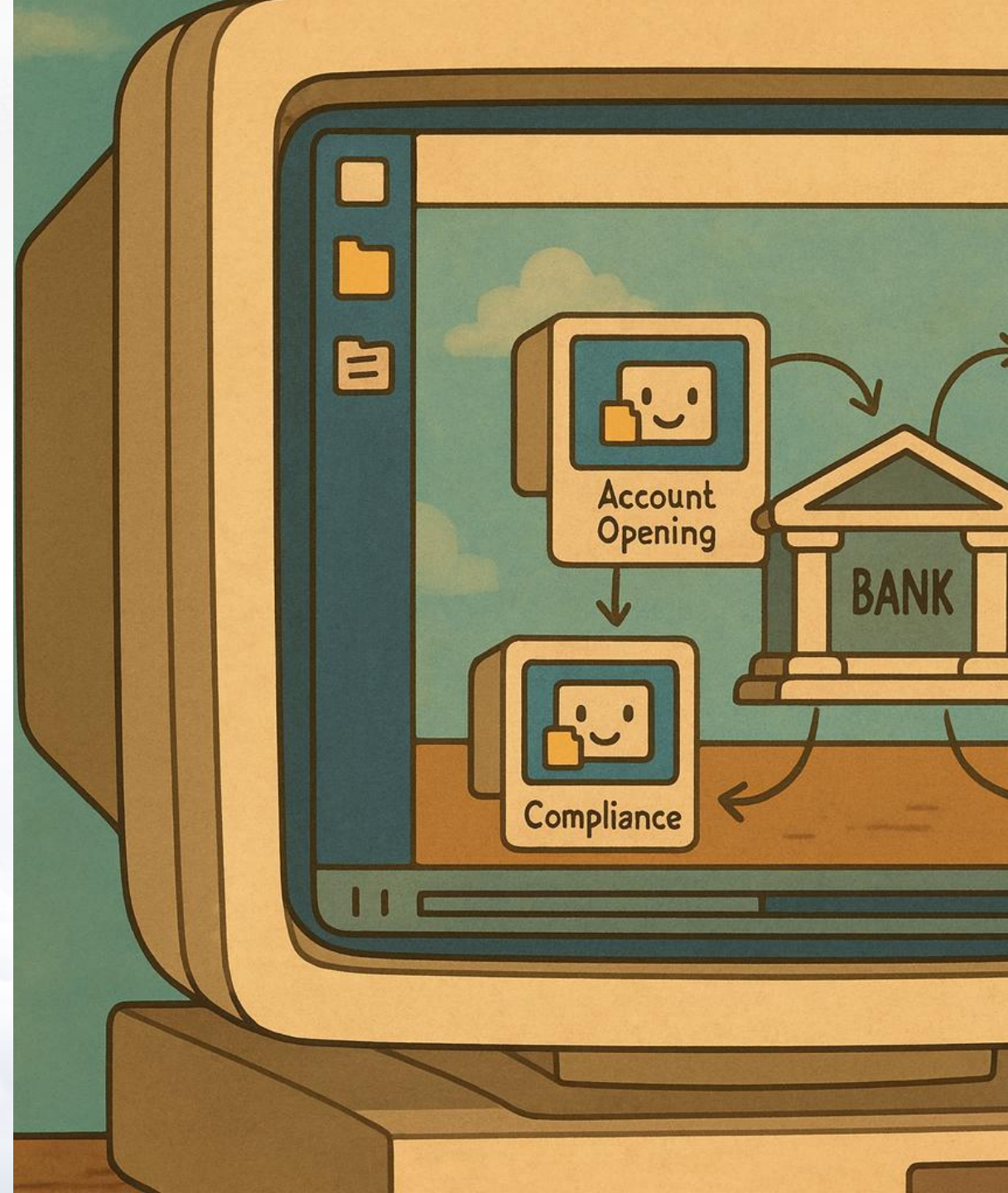




Financial Services and Insurance: Apps that matter

Disclaimer: This content is for Microsoft partner guidance only and not to be shared directly with customers.



What is an AI frontier firm

→ The Frontier Firm Defined

A next-generation company that fuses human judgment with AI agents to scale faster, work smarter, and unlock new value - powered by on-demand intelligence and human + agent hybrid teams.

💡 Core Characteristics

- **AI-operated, human-led:** Agents execute tasks; humans provide oversight.
- **Work Chart vs Org Chart:** Teams form around outcomes, not departments.
- **Agent boss mindset:** Every employee manages and collaborates with AI agents.

L Why This Matters Now

- 82% of leaders say 2025 is a pivotal year to rethink strategy and operations.*
- 80% expect agents to be integrated into their AI strategy within 12–18 months.*
- Frontier Firms outperform peers in productivity, innovation, and speed to value

"As AI democratizes access to expertise and intelligence, we'll see the rise of Intelligence Resources departments -much like how HR and IT evolved into core functions. These new divisions will be essential for managing the interplay between humans and AI agents, emerging as a critical source of competitive advantage in the AI-enabled enterprise."

-Karim R. Lakhani, Chair of the Digital, Data, and Design Institute at Harvard,

Microsoft Work Trends Index Report



AI Apps that matter:

Build AI Apps that drive Frontier Transformation

Business Impact

Delivers clear, measurable value by solving a real problem aligned with user needs and organizational goals: high-value use cases

Innovative

Leverages cutting-edge agentic technologies and real time data to solve complex problems in ways that are innovative, adaptive, and often predictive or autonomous

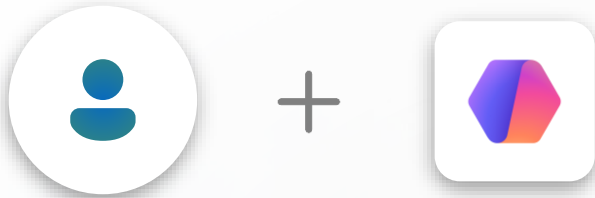
Enterprise-grade

Production-grade systems that are governed, secure, trusted, operationally stable, business-aligned, continuously optimized

Journey to the Frontier Firm

Phase 1

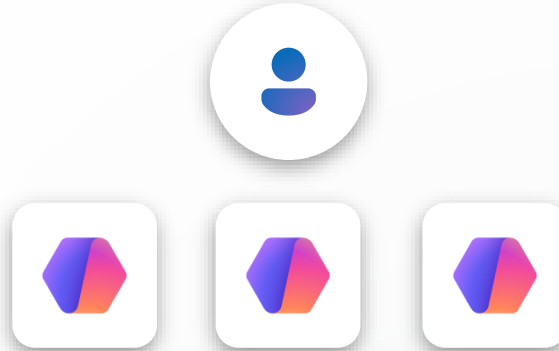
Human with assistant



Every employee has an AI assistant that helps them work better and faster

Phase 2

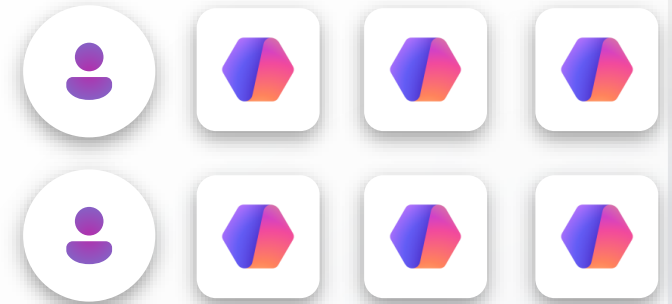
Human-led agents



Agents join teams as "digital colleagues," taking on specific tasks at human direction

Phase 3

Human-led, agent-operated



Humans set direction and agents run entire business processes and workflows, checking in as needed

Becoming Frontier

Success framework

Enrich employee experiences

Reinvent customer engagement

Reshape business processes

Bend the curve on innovation



Approach



Adopt Microsoft copilot



Build in Copilot Studio



Build on Microsoft Azure

Guardrails on how to work with the suggested processes

See the whole process



Map the end-to-end flow, actors, policies, and desired outcomes

Call out where agents could take actions

Decompose into "agent-able" jobs



Prioritize by value $\times$ feasibility $\times$ data readiness

Define success metrics and guardrails; timebox the first increment

Select the starting "slice"



Break into sub-processes with clear inputs, decisions, actions, outputs

Create a data map: sources, owners, contracts, latency, quality

Build an enterprise-ready agent



Design for extensibility: APIs/events, modular tools, config over code

Safety & compliance: identity/permissions, policy/RAG, tracing

Scale along the chain



Add/compose agents, automate handoffs, integrate with systems of record

Track ROI; harden to repeatable IP and reuse components

Ground rules: Agentic processes are a combination of multiple software solutions

FSI frontier solutions

Overall industry specifics:

- Augmentation-first due to risk/compliance: Human-in-the-loop by design; automate only low-risk, reversible steps; require explainability, audit trails, and approvals
- Ecosystem over custom build: Greater reliance on specialized third-party platforms (KYC/AML, fraud, core banking); agents compose and trigger via APIs rather than replace

1	Loan application processing	Collecting applicant data and documents, running KYC/AML and credit scoring, and routing to underwriting for an approve/decline with terms	
2	Mortgage processing	Collecting borrower and property data, verifying identity/income, performing credit/affordability and collateral checks to underwriting for approve/conditions/decline with closing disclosures	
3	Fraud & AML management	Monitoring transactions and accounts in real time with rules + ML, enriching with device/behavioural signals, and triaging alerts to cases for timely block/release decisions	
4	Claims processing	Taking FNOL to settlement by checking coverage, assessing liability, reviewing documentation, coordinating repair/replacement, and pursuing subrogation where applicable	
5	Market research & analytics	Analyze market data and news, economic indicators, earnings calls extracting key insights, building financial models, identify investment opportunities, and creating market and investment reports including tailored insights for clients	

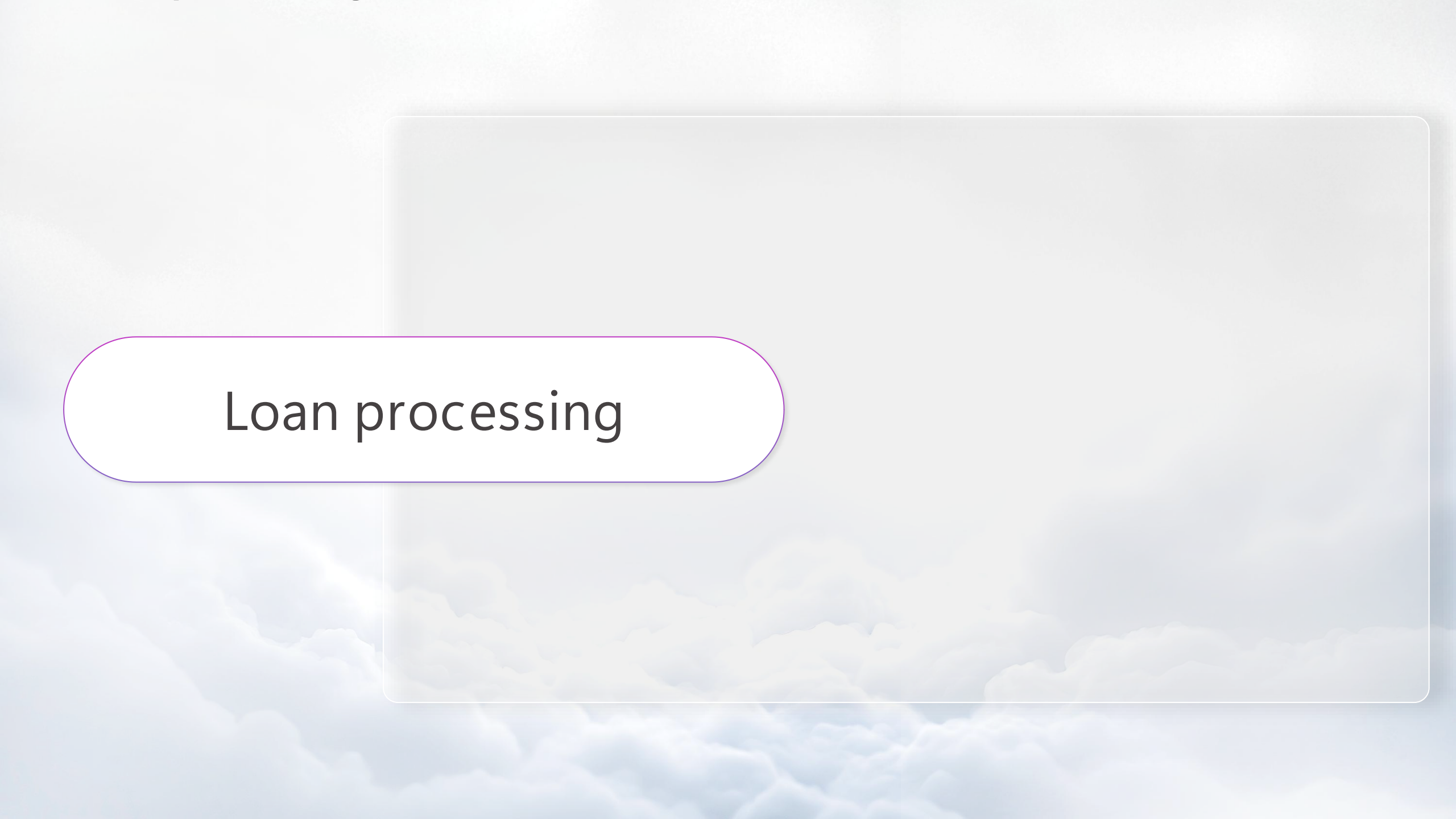
Banking focus

Insurance focus

Capital Markets

Key metrics relevant to the industry

- Manual-review rate & decision turnaround time: % of cases needing human review and time-to-decision (KYC, fraud, credit)
- Compliance/QA pass rate: disclosures/consents adherence, exception rate
- Fraud loss & false-positive rate: net loss per \$ volume and % good customers challenged/declined



Loan processing

1. Loan application processing

Loan processing is a core banking process that moves a customer from application to funded account while protecting the bank's risk posture and meeting regulatory obligations. Each application must be evaluated quickly for eligibility, affordability, and fraud/AML risk; missing items collected; decisions documented for audit; and, if approved, funds disbursed, and accounts set up with clear customer communications.

What - Process overview



Current challenges

- Fragmented intake across channels; incomplete or inconsistent documents slow reviews and create rework
- Manual data entry and duplicate checks across LOS, CRM, KYC/credit systems drive long cycle times
- Legacy rule/scorecard models lead to over/underwriting; explainability and policy alignment vary by analyst
- Case notes and credit memos drafted by hand - variable quality and heavy effort to prepare for exams/audits
- Siloed origination, underwriting, and ops teams; FIFO queues hide high-value or at-risk applications



Frontier solution

Process description

- Real-time intake triages each application by enriching with bank, device, employment, income, collateral, and KYC data; performs link analysis; assigns priority and a recommended action (approve, reject, conditional, escalate)
- Auto-compiled "credit file" evidence packs (timeline, entities, features used) generate draft credit memos and customer-ready communications; pushes updates to the loan origination system
- Orchestrated verifications via secure channels (document upload/e-sign, income & employment checks); outcomes recorded, and tasks closed/updated automatically
- One underwriting workspace with guided playbooks and policy retrieval to ensure consistent, compliant decisions

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Document processing agent** validates applications, performs OCR/structure, and opens conditions
- **Underwriting agent** routes work, deduplicates checks, and proposes approve/ conditional/ reject
- **Loan setup agent** assembles the credit memo, enforces policy & KYC/AML, handles disclosures/e-sign, and tracks condition clearing to booking
- **Responsible AI agent** that reviews the loan decisions to ensure fairness and equality

Why – Impact¹



Business Outcomes²

Cost savings:

- Shorter cycle times via automated intake, verification, and memo drafting: 30–50% faster underwriting and closing
- Fewer touchbacks/rework reduce ops effort and vendor spend: 25–40% lower cost per booked loan

Growth

- Higher pull-through from faster time-to-yes and risk-based decisions; better CX and conversion
- New products/segments (instant small-ticket, digital mortgage, BNPL) with controlled risk enable market expansion



Roles addressed

- Loan officers / originations ops
- Underwriters / credit analysts
- Closing / servicing operations & credit risk managers

How – Underlying foundation¹

Data to integrate

- Bureaus, bank/open-banking, income/employment, collateral
- KYC/AML, identity/device telemetry; LOS, CRM, pricing
- E-sign/docs, title/settlement, outcomes & performance

Technical patterns

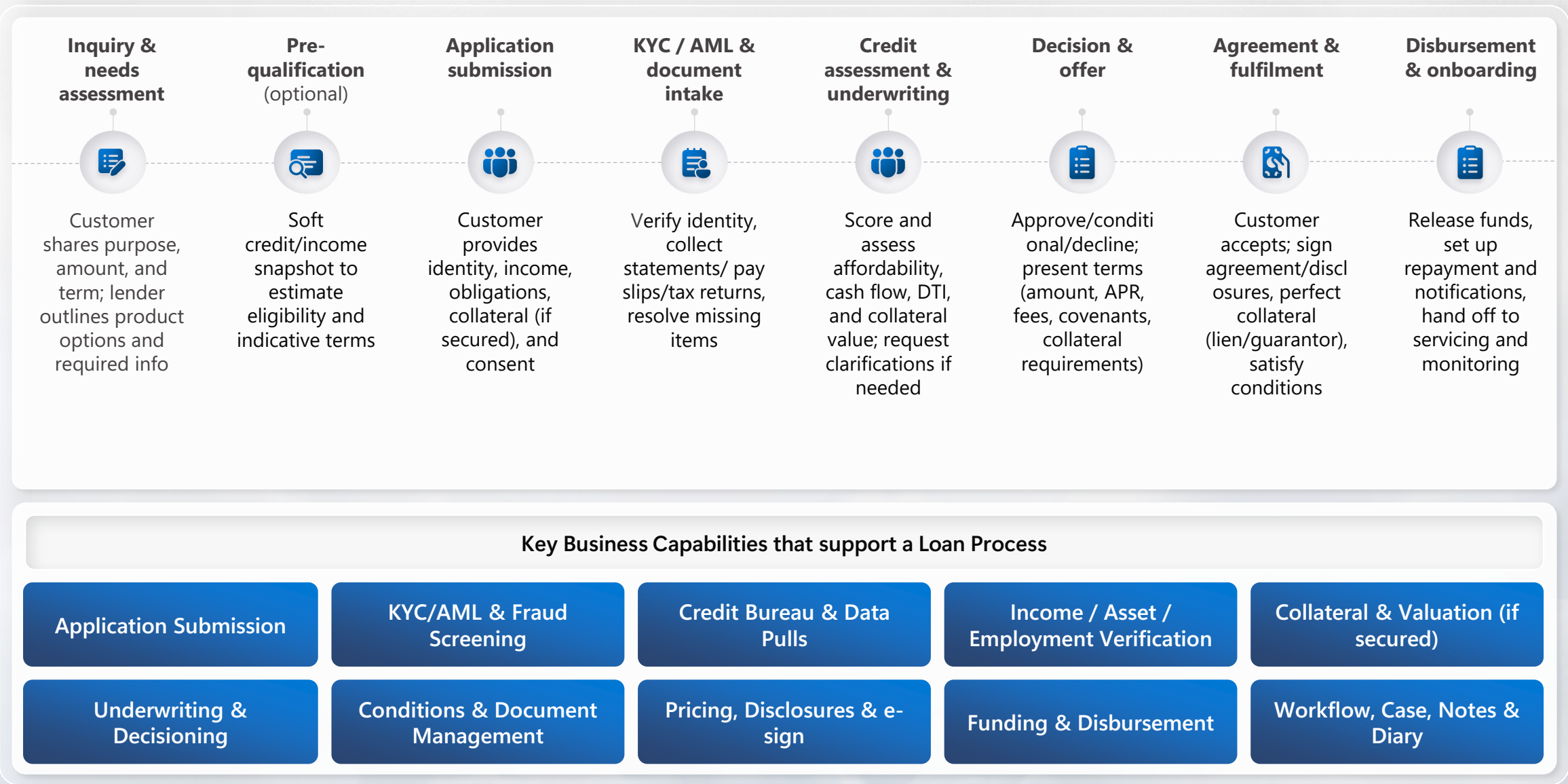
- Event-driven orchestration + real-time scoring, feature store & entity resolution
- RAG over credit policy/playbooks with tool-use into e-sign/disbursement
- Human-in-the-loop with auditable traces, model-risk governance

¹: All values related to „Future – AI as digital colleague“ state

²: ROI calculation performed in next step

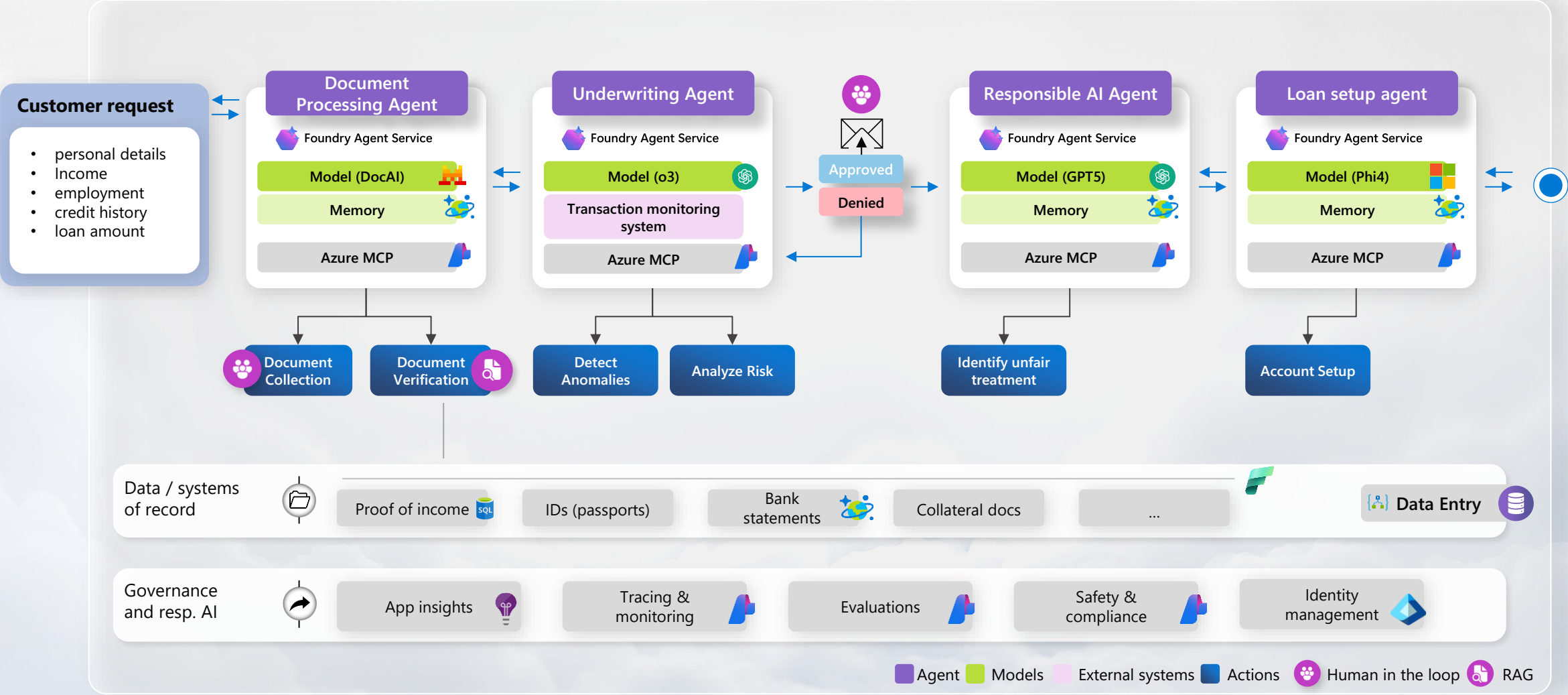
[Backbase | Fraud management and security](#)

1. What does the loan process look like?



1. Example Agentic Loan Processing

Loan application process



Mortgage processing

2. Mortgage processing

End-to-end mortgage application processing that automates document intake, verifies information, evaluates creditworthiness and affordability, and assembles the closing package. This shortens cycle time, reduces manual effort and errors, and gives applicants clear, timely updates.

What - Process overview



Current challenges

- Manual document chasing and rekeying extend turn times, increase error rates, and frustrate both applicants and loan staff
- Fragmented systems and handoffs cause duplicate work, inconsistent data, and frequent back-and-forth to reconcile discrepancies
- Policy interpretation varies by person, creating inconsistent decisions, avoidable exceptions, and costly rework during audits
- Slow exception handling and unclear ownership stall approvals, especially for edge cases that need coordinated review
- Heavy audit/compliance effort with limited traceability drives after-the-fact evidence gathering and remediation cycles



Frontier solution

Process description

- Applicant submits digital application; system pre-validates fields and requests consents
- Documents are classified, OCR'd, reconciled; missing or stale items are auto-requested
- Credit, income, identity, AML/sanctions, and property checks run with explainable results
- Underwriting proposes approve/conditional/reject with cited evidence and clear conditions
- Disclosures, e-sign, closing docs, and booking/funding are orchestrated with status updates

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Document processing agent:** Validates applications, classifies/extracts documents, identifies gaps, and reconciles inconsistencies
- **Underwriting agent:** Applies credit policy/affordability rules, runs KYC/AML/sanctions checks, compiles evidence, proposes decision
- **Loan setup agent:** Generates disclosures/closing package, coordinates e-sign, books loan in core systems, schedules funding tasks
- **Responsible AI agent** that reviews the loan decisions to ensure fairness and equality

Why – Impact¹



Business Outcomes²

Cost savings:

- 30–50% less manual handling
- 20–35% faster app-to-decision
- 10–20% fewer resubmissions

Growth

- 5–10% higher pull-through; faster cycle times and proactive updates lift CSAT/NPS



Roles addressed

- Fraud investigators and case analysts
- Loan officer
- Underwriter

How – Underlying foundation¹

Data to integrate

- Application & document streams, e-signature envelopes and disclosure statuses
- Credit/affordability & risk signals (bureau scores, employment/ income verification, KYC/AML, sanctions/watchlists)
- Property/valuation and product/pricing

Technical patterns

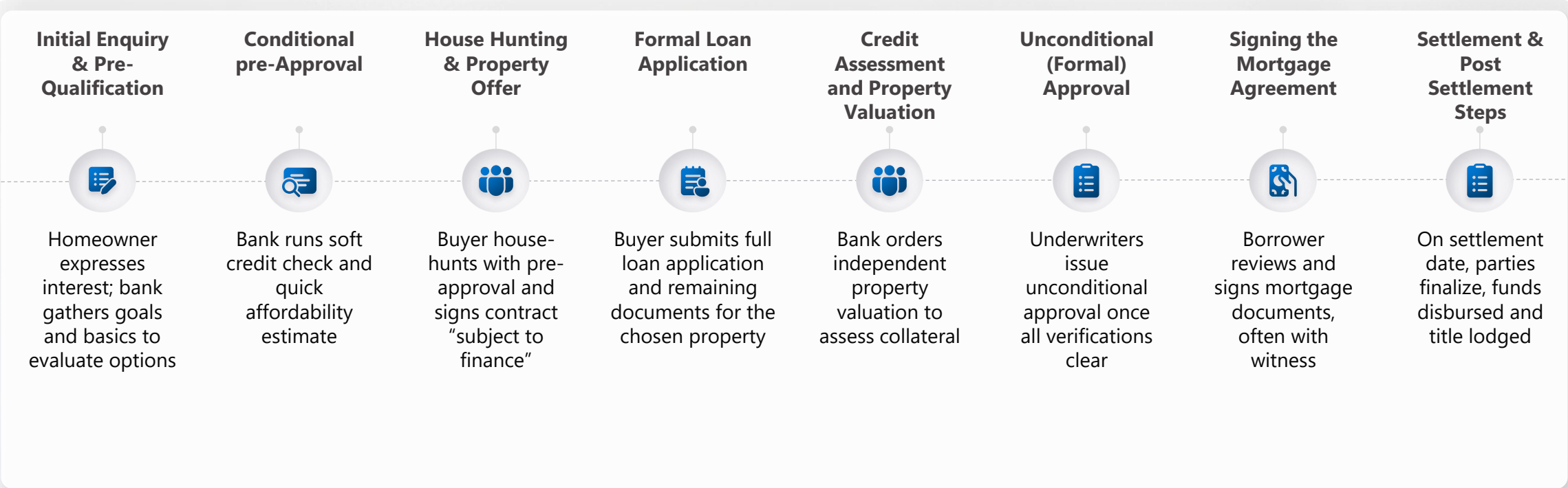
- Event-driven orchestration and stateful workflows
- Document understanding (classification, extraction, validation)
- Policy-aware decisioning with guardrails and audit trails

¹: All values related to „Future – AI as digital colleague“ state

²: ROI calculation performed in next step

2. What does the mortgage look like?

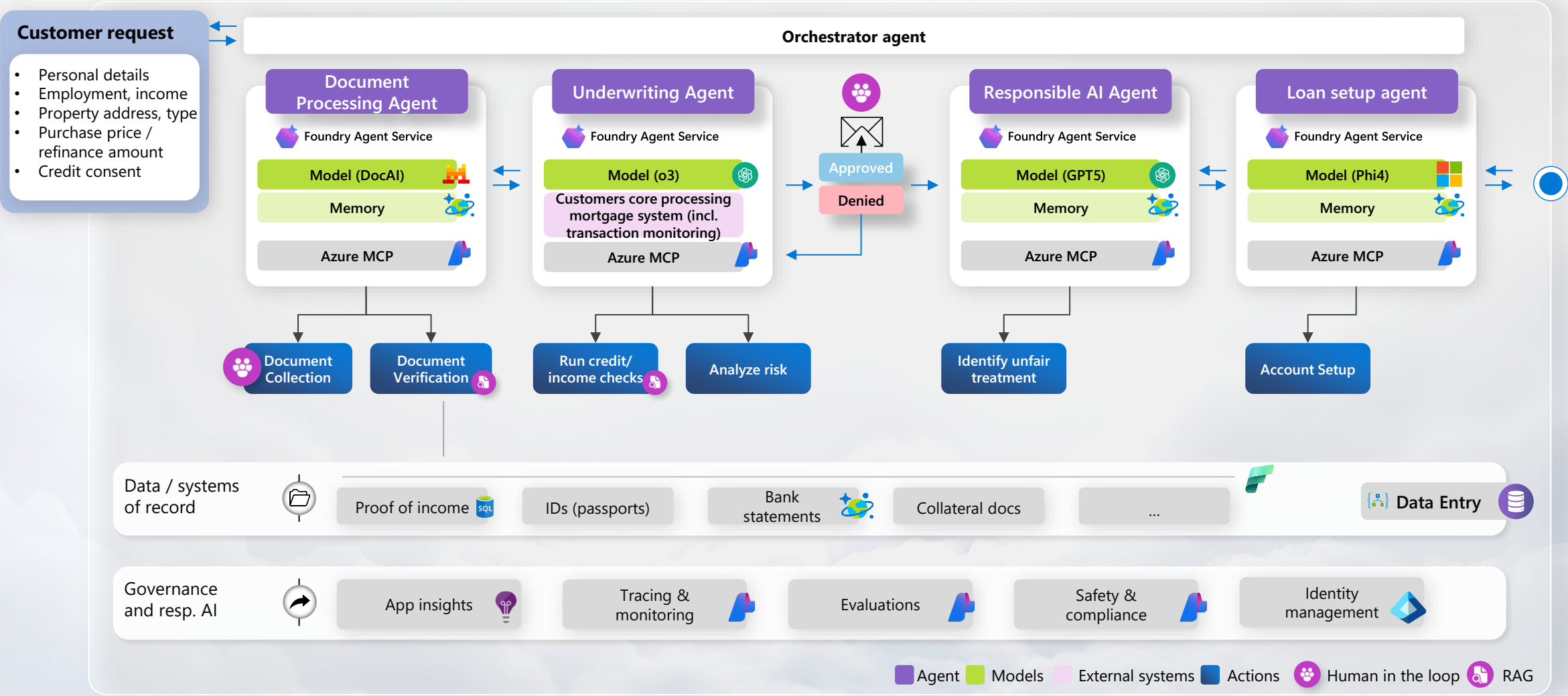
EXEMPLARY PROCESS - SIMPLIFIED



Key Business Capabilities that support a Loan Process

Mortgage application & pre-approval	Identity verification & AML/fraud checks	Credit bureau, income & asset verification	Property, appraisal, title & flood	Pricing & rate lock
Loan Estimate & Closing Disclosure orchestration	Underwriting & conditions management	Closing, recording & disbursement	Escrow (taxes/insurance) & mortgage insurance	Servicing setup, statements & communications

2. Example of agentic mortgage



Fraud management

3. Fraud & money laundering (AML) management

Fraud management protects customers and the bank across account opening, login, and payments by detecting and stopping identity theft, scams, and transaction abuse while meeting AML obligations. Each event is evaluated in real time for identity, device, behaviour, and network risk; evidence is retained for audit; cases are triaged for investigation and recovery, and customer communications are clear and timely.

What - Process overview



Current challenges

- Fragmented signals across channels and vendors; missing context drives high false positives and manual reviews
- Static rules decay and miss new scam patterns; duplicate alerts and siloed queues slow response
- Manual investigations and suspicious activity report (SAR) drafting are time-consuming; weak explainability hinders decisions and audits
- Limited linkage across accounts/devices/merchants; mule networks and ATO hard to identify
- Poor feedback loops from chargebacks and outcomes; models and controls aren't continuously improved



Frontier solution

Process description

- Real-time risk scoring enriches events with device, behavioural, geo, merchant, and KYC data; links entities to known networks
- Risk-based controls trigger step-up auth, delays, limits, or blocks; safe traffic flows without friction
- Alerts with auto-compiled "evidence packs" (timeline, entities, features used) are prioritized and routed to the right team
- Guided investigations retrieve policies, similar cases, and checklists; case notes and actions are captured automatically
- Automated drafting for SARs/claims/representments; outcomes feed back into features, rules, and models for continuous learning

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Rules & policy agent:** Author/update monitoring rules and explain policy logic in plain language
- **Investigator triage agent:** Review suspicious accounts/transactions and propose approve/hold/escalate pre-human
- **Link analysis & narrative agent:** Uncover case linkages, summarize evidence, and draft SAR narratives
- **Guardrail QnA agent:** Answer regulatory geo questions and surface supporting external evidence for analysts

Why – Impact¹



Business Outcomes²

Cost savings:

- Fewer manual reviews via better precision and deduplication
- Lower operational and vendor costs through automation and targeted controls; reduced post-loss handling

Growth

- Higher approval rates and fewer false declines improve conversion and CX across cards
- Safer instant payments and new products (P2P, wallets, cross-border) with controlled fraud loss enable expansion



Roles addressed

- Fraud investigators and case analysts
- AML/compliance officers and sanctions teams
- Security operations managers

How – Underlying foundation¹

Data to integrate

- Real-time transactions (card, ACH/SEPA, wires, RTP), device/browser telemetry, session and channel logs
- Customer/account/KYC/CRM, sanctions/PEP/watchlists, merchant/acquirer data
- Case management actions

Technical patterns

- Event-driven streaming with real-time scoring; feature store, entity resolution, and graph analytics for network risk
- RAG over policies/runbooks; human-in-the-loop investigations with auditable traces and challenger monitoring

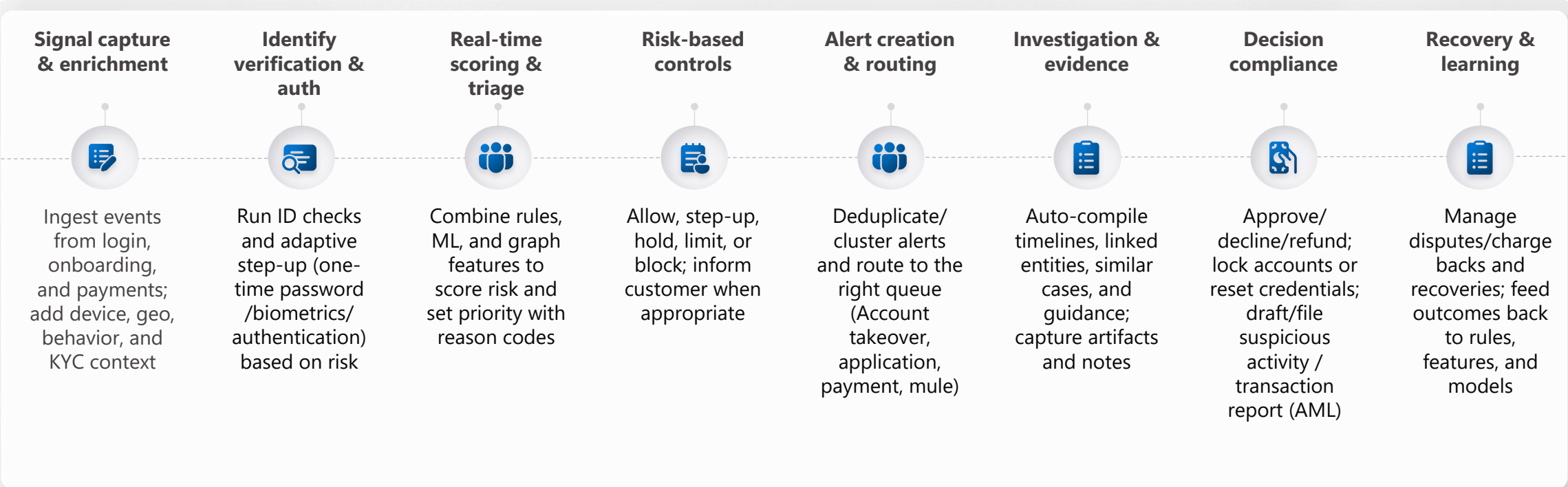
¹: All values related to „Future – AI as digital colleague“ state

²: ROI calculation performed in next step

[Backbase | Fraud management and security](#)

3. What does the fraud management process look like?

EXEMPLARY PROCESS - SIMPLIFIED

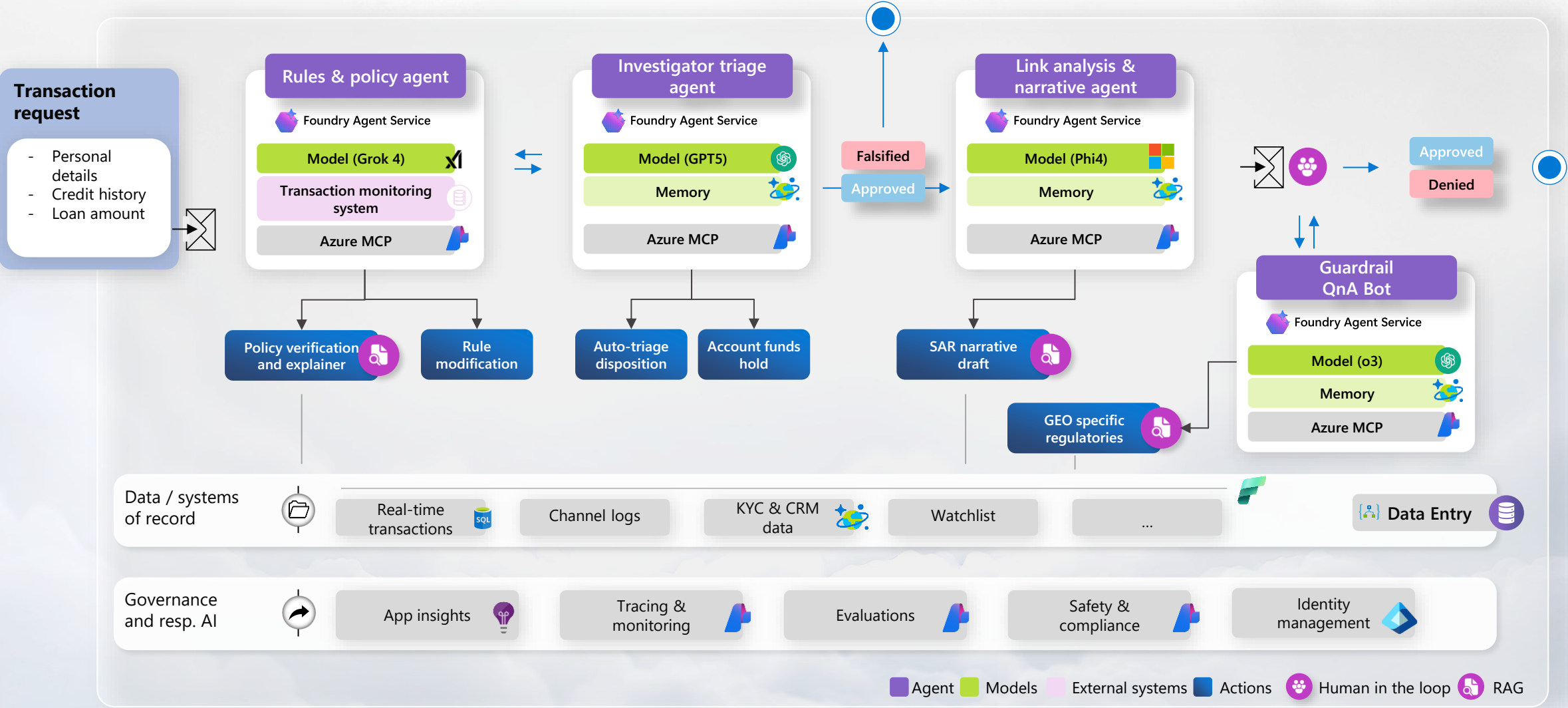


Key Business Capabilities that support a Fraud Management Process



3. Example of Agentic Fraud Management and Money Laundering

Exemplary process for B2C



Claims processing

4. Claims processing

Claims processing moves a policyholder from first notice of loss (FNOL) to settlement and recovery while protecting the insurer’s loss ratio, compliance posture, and customer trust. Each claim is evaluated for coverage, liability, severity, and fraud risk; documentation is captured for audit; repairs/medical services are coordinated; payments are disbursed; and subrogation/salvage pursued—with clear communications throughout.

What - Process overview



Current challenges

- Fragmented FNOL across phone, apps, and partners; missing or inconsistent documents/photos delay triage
- Manual coverage verification and policy interpretation drive rework and variable decisions
- Disconnected vendors (repair, appraisal, medical, legal) and handoffs extend cycle times
- Inconsistent reserving and estimate quality create leakage and post-payment corrections
- Reactive fraud detection with limited link analysis; high false positives and missed rings



Frontier solution

Process description

- Real-time FNOL triage enriches events with policy, telematics, photos/video, geo/weather, and loss history; links parties and assets
- Auto-drafted coverage determination and liability hypothesis propose next best path (straight-through, desk, field, SIU)
- Orchestrated verifications (police report, appraisal/AVM, medical bill review, title/salvage) run via secure channels; outcomes posted to the claims system
- Auto-compiled “claim file” evidence pack with reserve suggestion, tasks, and customer-ready communications
- One adjuster workspace with guided playbooks through estimate, negotiation, settlement, subrogation, and recovery

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **FNOL (first notice of loss) intake & document agent** captures loss, validates policyholder, structures forms/photos, opens claim
- **Triage & coverage agent** interprets policy/endorsements, scores severity and fraud risk, routes STP/desk/field/SIU, sets reserves
- **Assessment, settlement, recovery agent:** Reviews damage, coordinates vendors/adjusters, drafts estimates/settlements, issues payments, handles subrogation/salvage
- **Responsible ai & policy guardrails agent:** Enforces fair-claims and regulatory rules via RAG, deduplicates fraud alerts, monitors exceptions and model risk with audit trails

Why – Impact¹



Business Outcomes²

Cost savings:

- 25%-45% faster cycle times via automated intake, verifications, and drafting
- Reduced leakage through guided decisions and estimate/medical controls: 10–20% lower indemnity leakage

Growth

- Better CX from faster, transparent settlements improves retention/NPS and lowers re-contacts
- Digital FNOL and straight-through processing enable new products/channels and partner ecosystems



Roles addressed

- Claim handlers, desk & field appraisers
- SIU investigators, subrogation & recovery specialists

How – Underlying foundation¹

Data to integrate

- Policy admin/underwriting, core claims, CRM, communications and consent
- Photos/video, telematics/dashcam, repair estimates (CCC/Audatex), parts/labor databases, salvage data
- Payments/reserves, legal and subrogation outcomes, ISO Claim Search, third-party intel sources

Technical patterns

- Event-driven orchestration with real-time scoring; feature store + entity resolution and graph analytics for fraud/subrogation
- Computer vision for damage assessment; LLM doc understanding for forms/bills/letters; tool-use into claims, repair, medical, and payment systems

¹: All values related to „Future – AI as digital colleague“ state

²: ROI calculation performed in next step

4. What does the claims process look like?

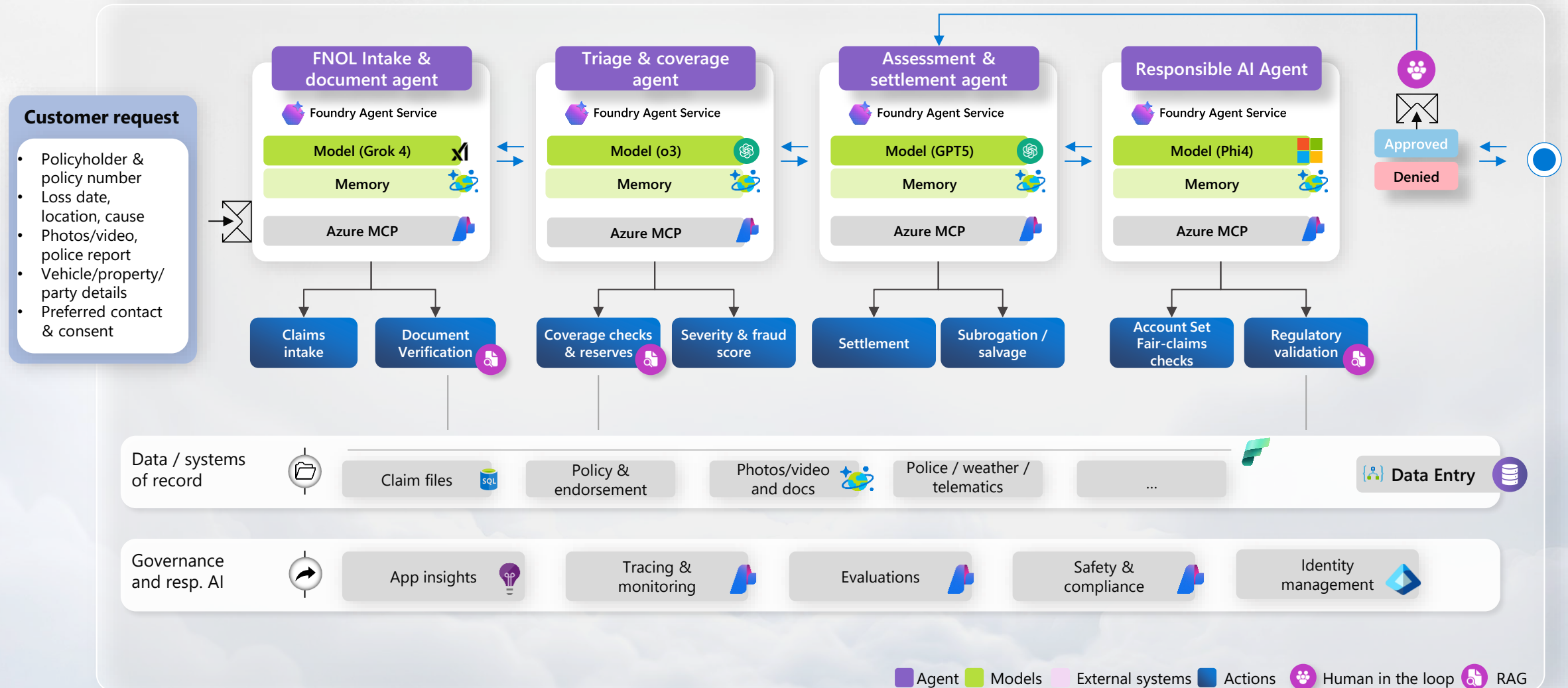
EXEMPLARY PROCESS - SIMPLIFIED



Key Business Capabilities that support a Claims Process

Claims Submission	Adjudication/ Case Management	Calendar, Notes, Diary	Workflow rules	Recoveries
Payments	Reserving	Fraud Scoring	Vendor/Provider Management	Litigation Management

4. Example agentic claims processing



4. Agentic Claims Processing



Contoso AI Claims

Live Dashboard

Agent Demos

- Claim Assessor
- Policy Checker
- Risk Analyst
- Communication Ag...

Workflow Demo

Documents

- Policy Documents
- Document Manage...
- Index Management
- More

Dashboard

Claims Processed

2,847

Processing efficiency up ↗
Claims processed this month

Active Policies

45,678

Strong policy retention ♥
Policies currently in force

Fraud Detection Rate

12.8%

AI detection improving ⬆
Suspicious claims identified

Average Settlement

\$8,450

Cost control effective ↘
Per claim settlement amount

Policies

☐

Policy Name

Policy Type

Status

Coverage

Limit

☐	Comprehensive Auto Policy 🔗	Auto Insurance	● Active	Full Coverage	\$100,000	⋮
☐	Commercial Auto Policy 🔗	Commercial	● Active	Business Fleet	\$500,000	⋮
☐	High Value Vehicle Policy 🔗	Premium Auto	● Active	Luxury Vehicles	\$250,000	⋮
☐	Liability Only Policy 🔗	Basic Auto	● Active	Minimum Coverage	\$50,000	⋮
☐	Motorcycle Policy 🔗	Motorcycle	● Active	Two-Wheeler	\$75,000	⋮

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Rows per page 10 Page 1 of 1

AI Agent Demos

Experience our multi-agent insurance claim processing system

Multi-Agent Workflow 🔗

Experience how AI agents collaborate to process insurance claims in real-time

Full Demo ▶ Test

Claim Assessor 🔗

Analyzes claim details and determines initial assessment with risk scoring

Assessment ▶ Test

Policy Checker 🔗

Validates policy coverage and verifies terms for claim eligibility

Validation ▶ Test

Risk Analyst 🔗

Evaluates fraud indicators and calculates comprehensive risk factors

Risk Analysis ▶ Test

Communication Agent 🔗

Generates personalized customer communications and notifications

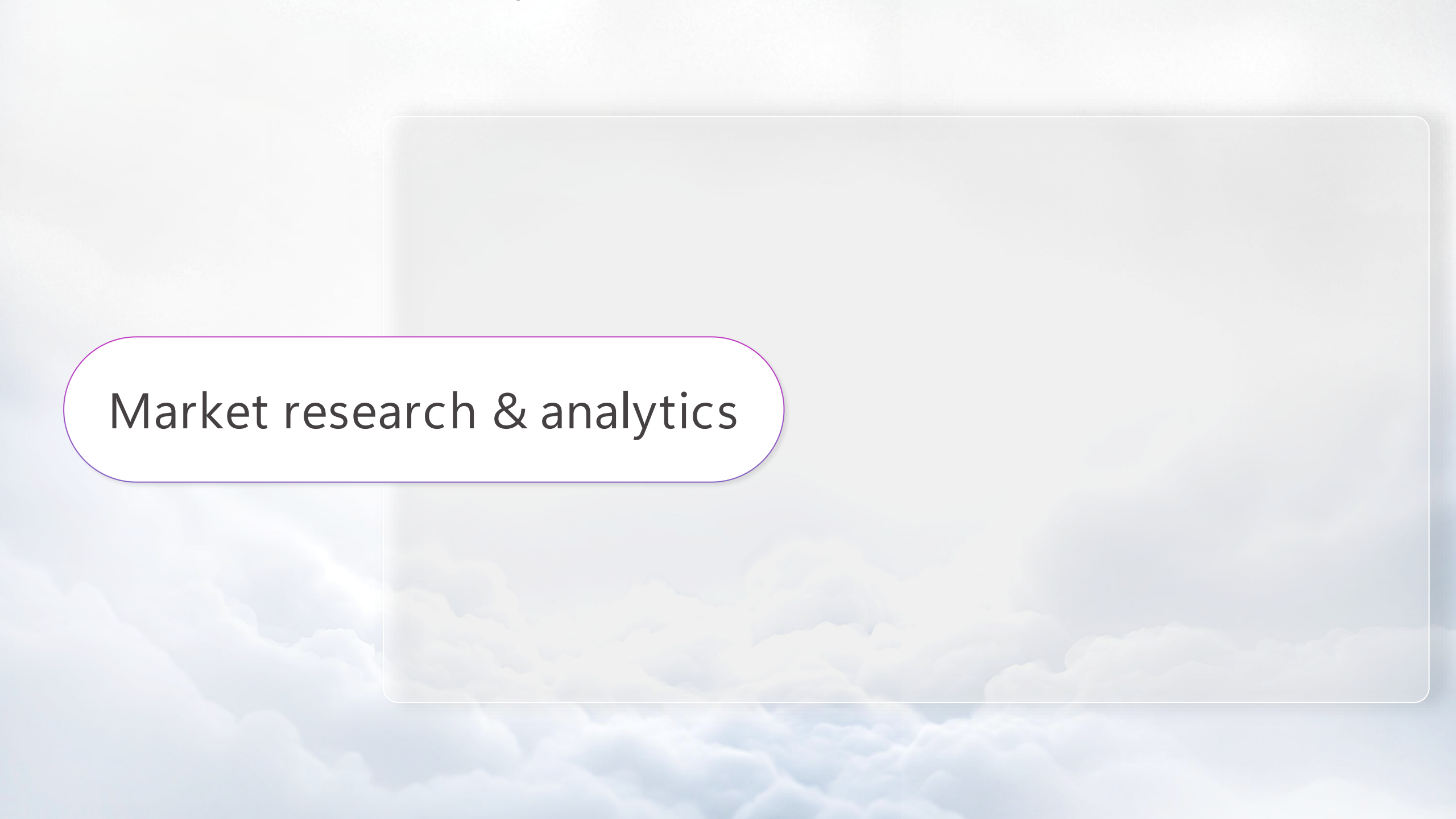
Communication ▶ Test

Each agent demo includes sample data and connects to the live backend API

Light mode

localhost:3000/#

N



Market research & analytics

Market research & analytics (investment research report generation)

Research and analytics in capital markets involve the continuous generation of insights from structured and unstructured data to support investment decisions, market commentary, and client advisory. Analysts must synthesize vast volumes of data from earnings calls, regulatory filings, market news, and internal systems - often under tight deadlines and volatile conditions

What - Process overview



Current challenges

Legacy systems, siloed data, and manual workflows hinder speed and accuracy

- Siloed data: Research analysts struggle to access and synthesize data across platforms
- Manual workflows: Time-consuming data compilation and report generation
- Delayed insights: Lag in decision-making due to lack of real-time analytics
- Tool fragmentation: Navigating multiple platforms reduces productivity
- Compliance bottlenecks: Manual review processes slow down publication



Frontier solution

Process description

- AI-powered multi-agent systems transforming this landscape by automating data collection, analysis, and reporting, enabling analysts to focus on high-value insights and strategic recommendations
- Agent Flows: Predefined sequences automate repetitive tasks like report drafting or earnings call summarization
- Integration: Agents connect to Research Management Systems (RMS), market data feeds, and productivity tools like Microsoft 365 Copilot
- Orchestration: A lead deterministic agent delegates tasks in order to specialized sub-agents (e.g., data extraction, modeling, summarization)
- Human-on-the-loop: Research analysts validate, refine, and approve outputs, ensuring accuracy and compliance

Solution description

Agents collaborate via orchestrator agent and humans in the loop:

- **Research agent** gathers and synthesizes data from internal and external sources
- **Analyst agent** performs deep reasoning, financial modeling, and scenario analysis
- **Specialized sub-agents** (e.g., data extraction, modeling, earnings call summarization)
- **Collaboration agent** coordinates workflows and integrates feedback loops
- **Compliance agent** reviews content for regulatory alignment and flags issues
- **Presentation agent** converts insights into visualizations and client-ready formats
- **Orchestration agent** deterministically follows the business process call the other agents

Why – Impact



Business Outcomes

Business and Market Agility

- Faster time-to-insight and enhanced decision-making agility

Time Savings:

- 80% time savings on data collection and 50% on analysis (Moody's)

Improved compliance

- Improved compliance and reduced risk through automated review

Presentation generation

- 50% reduction in presentation generation and client meeting prep time



Roles addressed

- Research Analysts
- Investment Analysts
- Financial Analysts
- Chief Investment Officer
- Data Scientists & Quant Researchers
- Compliance Officers

How – Underlying foundation

Data integration

- Market data feeds
- Internal Research Management Systems (RMS) and CRM systems
- Unstructured data (e.g. earning calls, filings, news)
- APIs and data lakes for real-time access...

Technical patterns

- Event-driven orchestration: agents respond to triggers like earnings releases or market events
- Conversational interfaces for analysts to use natural language to query data and generate insights
- Human-in-the-loop: analysts review and approve agent-generated outputs...

What does the investment research report process look like?

EXEMPLARY PROCESS - SIMPLIFIED



Key Business Capabilities that support the research & analytics processes



Customer example for agentic investment research processes

Customer & Challenge

- Struggled to synthesize external market data with internal portfolio insights and research
- Investment hypotheses depended on individual judgement; limited tooling to connect public market signals to private company performance
- Data was siloed and unstructured, with no cross-department benchmarking capability

Solution

Agentic retrieval with an LLM that decomposes complex questions into focused, parallel sub-queries across third-party structured data and internal portfolio sources

Surfaces hidden patterns by fusing external market signals with proprietary portfolio information

Merges results into a single, grounded answer so teams get fast, complex responses enriched with their own content

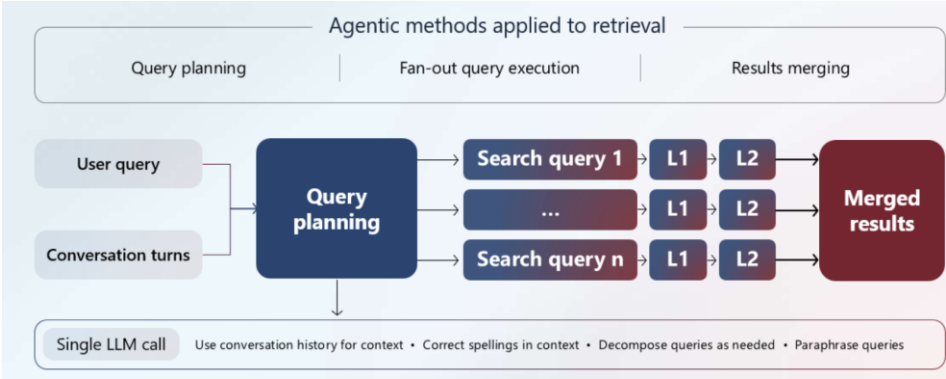
Business Outcome

Improved investment insight generation through large language model recognition across structured and unstructured data

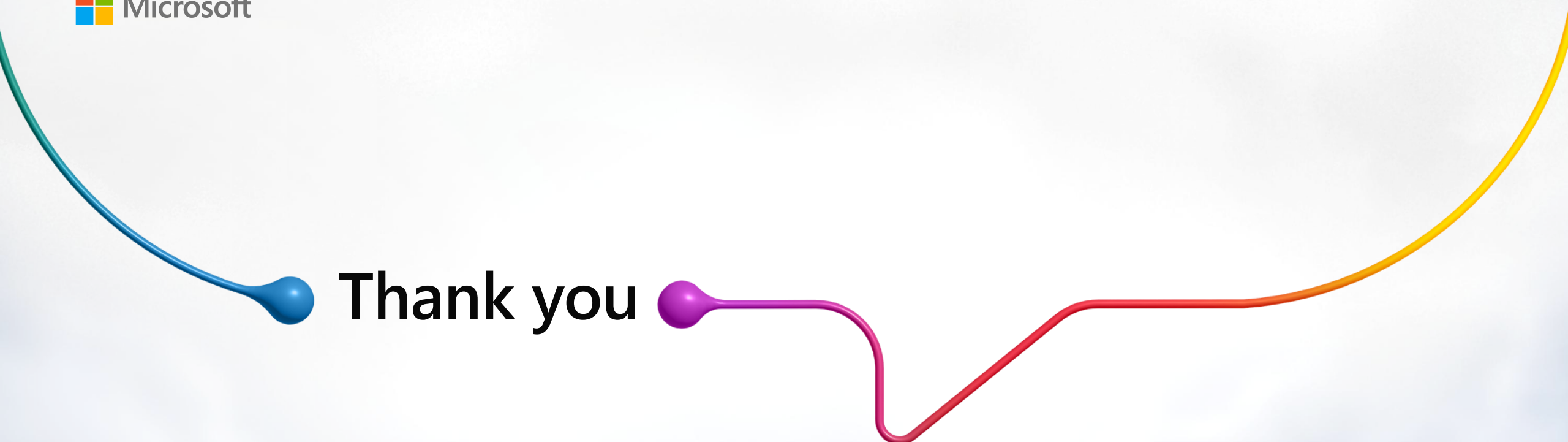
Faster time to market, results that used to take 10's of minutes to run, are now done in 10's of seconds

Reduced reliance on manual analysis, accelerating due diligence workflows and improving consistency in hypothesis validation

Architecture



Component	Service	Role
LLM	Azure OpenAI	Creates subqueries from conversation context and later uses grounding data for answer generation
Knowledge agent	Azure AI Search	Orchestrates the pipeline, connecting to your LLM and managing query parameters
Search index	Azure AI Search	Stores your searchable content (text and vectors) with semantic configuration
Semantic ranker	Azure AI Search	Required component that reranks results for relevance (L2 reranking)
Knowledge source	Azure AI Search	Wraps the search index with properties pertaining to knowledge agent usage

A decorative line with spheres. It starts as a blue line on the left, curves down to a blue sphere, then continues as a purple line to a purple sphere. From the purple sphere, it goes right, then down to a red line, then up to a red line, then right to a yellow line, and finally curves up to the top right corner.

Thank you